Analysis of OPTN/UNOS registry suggests the number of HLA matches and not mismatches is a stronger independent predictor of kidney transplant survival

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HLA matching and mismatching, while inversely related, are not exact opposites. Here we determined the independent effects of HLA matching and mismatching on outcomes in deceased donor kidney transplant recipients. The United Network for Organ Sharing database (1995–2012) was utilized and analyzed for delayed graft function, one-year acute rejection, and death-censored graft survival using combined multivariable models including HLA matching and mismatching. Sensitivity analyses were performed using the subgroup of deceased donor kidney transplant patients after 2003 with more uniform HLA nomenclature and resampling analyses using bootstrapping on complete data available from 96,236 recipients. Individually, both HLA matching and mismatching showed significant associations with graft survival. Adjusting the model to take into account both matching and mismatching simultaneously, the degree of HLA mismatching lost significance while matching continued to have a significant prediction for delayed graft function, the one-year acute rejection rate, and graft survival. Sensitivity analyses and bootstrapping showed similar results for all studied outcomes. Thus, analysis of this large cohort demonstrates the apparent greater association of HLA matching over HLA mismatching on both early allograft events as well as graft survival. Future analyses should preferentially utilize HLA matching as a covariate over mismatching for accurately reflecting impact on graft outcomes.

Kidney International (2018) **93,** 482–490; http://dx.doi.org/10.1016/j.kint.2017.07.016

KEYWORDS: acute rejection; delayed graft function; graft survival; HLA-matching; kidney transplantation; UNOS/OPTN

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Received 17 March 2017; revised 20 June 2017; accepted 20 July 2017; published online 5 October 2017

Increased human leukocyte antigen (HLA)-matching between donor and recipient has been repeatedly associated with improved graft survival in kidney transplantation.[1–4] This has led to the development of the Organ Procurement and Transplantation Network's (OPTN) kidney allocation policy of mandatory sharing for zero antigen-mismatched kidneys in patients with calculated panel reactive antibody >20%, and up to 2 points are assigned for a zero HLA-DR mismatch.[5] In spite of these policies, immune-related inflammation and progressive graft dysfunction remain the most important causes of long-term graft loss in single-center studies in the era of modern immunosuppression.[6] Large database studies affirm that the additive signal for the hazard of graft-loss proportional to levels of HLA mismatches is still maintained.[3,4] Others have argued that simply adding up the number of HLA-mismatched alleles while allotting organs adds little benefit given the present limitations in the donor pool and the efficacy of current immunosuppressive therapies.[7–9] Meanwhile, observational, biomarker-based and interventional research studies continue to use HLA matching or HLA mismatching as predictor variables for graft outcomes.

Many publications refer to HLA matching and HLA mismatching interchangeably. The absence of a mismatch is strongly and inversely correlated with the presence of a match. However, due to the occurrence of allelic duplication at the A, B, and DR loci, and the existence of alleles that are not identifiable by the current polymerase chain reaction probes/primers, the two terms are not synonymous. An illustrative example is shown in Figure 1. The hypothetical recipient would be 0-mismatched with each of the 3 potential donors shown here; yet donors 1, 2, and 3 would be 3-, 5- and 6-antigen matched with this same recipient, respectively. Independent of levels of HLA-mismatching, non-HLA genetic loci have been implicated in allorecognition[10] and associated with the recipient alloimmune response.[11–13] Hence, we hypothesized that HLA matching may be a surrogate for increased minor *loci* similarities that are not entirely mirrored by the number of HLA mismatches.

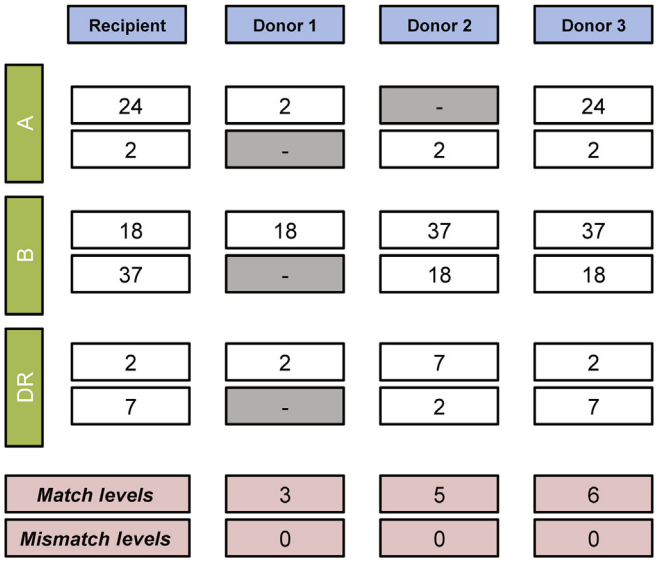


Figure 1 | Illustrative example of different levels of human leukocyte antigen (HLA)-matching within the same level of mismatch. In the United Network for Organ Sharing/Organ Procurement and Transplantation Network database, recipient 1 would have 0 mismatches (A, B, and antigen D-related [DR] loci) with each of the donors 1, 2, and 3. On the other hand, recipient 1 would be 3-, 5-, and 6-matched with donors 1, 2, and 3, respectively. Similarly, within each level of mismatch between donor and recipient, several different match levels are possible and vice versa.

Owing to the reporting of matches and mismatches in the United Network for Organ Sharing (UNOS), we tested the hypothesis that HLA matching has more significant correlation with allograft outcomes than HLA mismatching does. To examine this, we retrospectively examined the UNOS database including all deceased-donor kidney transplants from 1995 to 2012.

RESULTS

Study population (1995–2012)

A total of 96,236 deceased-donor kidney transplants were included based on complete available data (Figure 2). The mean follow-up of the entire cohort was 1482.5 days (median: 1105 [interquartile range: 382, 3163]). We defined 7 levels of HLA matches and mismatches (0–6 in both categories). Supplementary Tables S1A and S1B describe clinicodemographic characteristics within each level of HLA matching and HLA mismatching. As shown in Supplementary Figure S1, the distributions of transplant recipients of deceased donor kidney within the UNOS cohort by levels of HLA matching and HLA mismatching are not exact inverses. The matrix distribution of patients according to their levels of HLA matching and HLA mismatching in the entire cohort (Table 1) confirms that significant proportions of transplant recipients studied at each level of mismatch had discordant levels of matches in all categories (i.e., where number of matches was not equal to 6 – number of mismatches). The maximum concordance, was observed between 1-match and 5-mismatch (83.8%) and minimum concordance was observed between 0-mismatch and 6-match (33.4%) (Table 1, bold values).

HLA-match level has greater association with DGF than the HLA-mismatch level

To determine whether HLA matching or HLA mismatching demonstrated greater association with delayed graft function (DGF), we evaluated the effects of HLA matching and HLA mismatching on DGF risk (see Materials and Methods). We hypothesized that early alloimmune inflammation contributes to or prolongs DGF and would in turn be influenced by the level of HLA matching or HLA mismatching.[14,15] When individually analyzed in separate multivariate models, both matching and mismatching had significant associations with DGF, relaying the role of early alloinflammation in DGF (Figure 3a and b). However, when both match and mismatch were analyzed in a combined model, HLA-match levels had clear and significant association with DGF while HLA-mismatch levels had no impact (Figure 3c and d). In the sensitivity analyses, we generated similar logistic regression models for the subset from 2003 to 2012 with more homogenous HLA nomenclature (n = 70,759; see Materials and Methods). The mean follow-up of this 2003 cohort was 1072.8 days (median: 926 [interquartile range: 362, 1658]). The relationship between HLA-match levels and DGF, and the absence of significant association with HLA-mismatch levels, were consistent when the cohort after 2003 was separately examined (Figure 3e and f). Resampling analysis using bootstrapping technique with 10,000 iterations with resubstitution revealed similar results.

HLA-match level has greater association with 1yr-AR than the HLA mismatch level

Next, we examined the relationship of HLA matching and HLA mismatching on the outcome of clinical acute rejection within 1 year posttransplantation (1yr-AR). We used the data regarding early clinical 1yr-AR that is reported within the UNOS database. Individually both HLA matching and HLA mismatching associated significantly with 1yr-AR as an outcome (Figure 4a and b). When we input both in the same adjusted model (see Materials and Methods), we observed again that HLA-match levels were significantly predictive of 1yr-AR while HLA mismatching lost significance (Figure 4c and d). This effect of HLA matching levels on 1yr-AR and the absence of significant association with HLA-mismatching levels were consistent when the cohort after 2003 was separately examined (Figure 4e and f). Resampling analysis using bootstrapping technique with 10,000 iterations with resubstitution yielded similar results.

Sensitivity analysis. Due to the known significant impact of DGF on the risk of 1yr-AR (adjusted hazard ratio: 1.79; 95% confidence interval: 1.71–1.89 in the UNOS cohort; $P < 0.001$), and to further examine the interaction of DGF with HLA matching on the risk of 1yr-AR, we studied 1yr-AR risk in models in patients without DGF. In patients without DGF, HLA matching, and not HLA mismatching, had a significant effect on 1yr-AR risk in a combined model

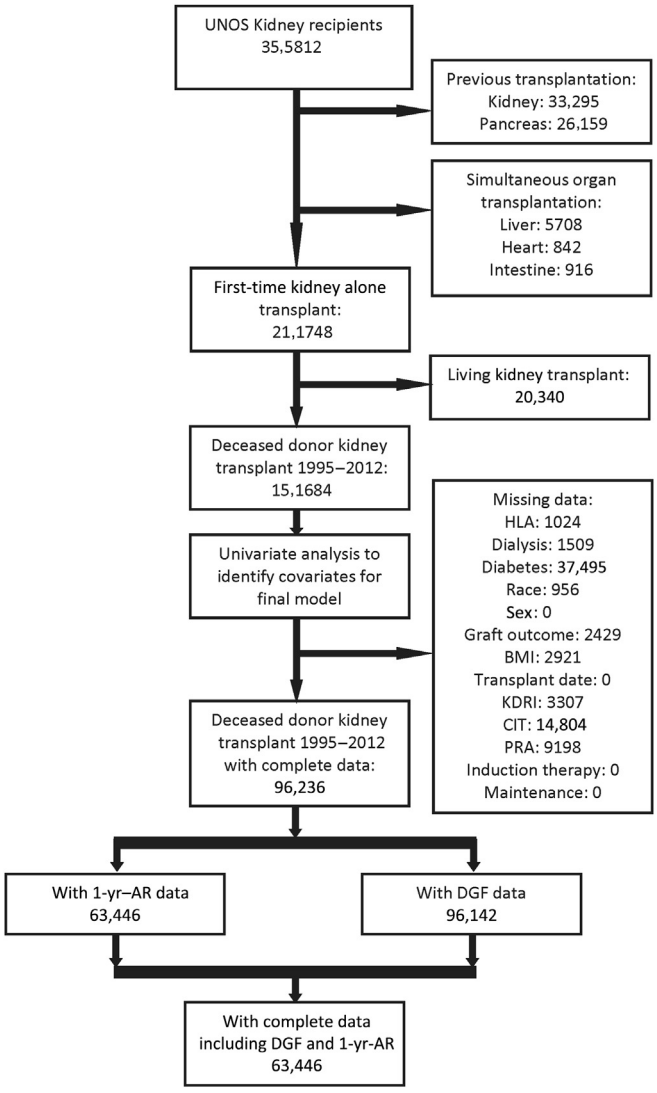


Figure 2 | Flow diagram of deceased donor kidney transplants performed from 1995 to 2012 used for analysis. The flow diagram shows the transplant recipients of deceased donor kidneys between 1995 and 2012 that were used for analysis of human leukocyte antigen (HLA)–matching and HLA-mismatching levels and allograft outcomes. Simultaneous nonkidney solid organ transplants, kidney-pancreas transplants, prior kidney transplants, and living-donor kidney transplants were excluded from univariate analysis. Only transplant recipients of a deceased donor kidney with complete data for covariates were included in the final models. 1-yr AR, clinical acute rejection within 1 year posttransplantation; BMI, body mass index; CIT, cold ischemia time; DGF, delayed graft function; KDRI, kidney donor risk index; PRA, panel reactive antibody; UNOS, United Network for Organ Sharing.

($P = 0.01$ vs 0.074, respectively), with similar results in the 2003 to 2012 cohort (Supplementary Table S2).

HLA-match level has greater association with graft loss than the HLA-mismatch level

Table 2 shows clinical and demographic factors associated with graft survival in univariate analysis. There were a total of 13,295 events of death-censored graft loss in the cohort within 10 years of follow-up. In separate Cox models, decreasing levels of HLA matching and increasing levels of HLA mismatching were associated with corresponding reductions in 10-year death-censored graft survival (DCGS) (Figure 5a and b). These findings are consistent with prior UNOS/OPTN database analyses. As a sensitivity analysis, we generated similar Cox-regression models for the subset from the 2003 to 2012 cohort. Here again, when considered individually, every level of HLA matching and every level of HLA-mismatching was associated with graft survival in a step-wise fashion (Supplementary Table S3).

Analogously, we explored the association between HLA-matching and DCGS in combined models. We observed that while increasing levels of HLA matching remained significantly associated with a decreasing hazard of graft loss ($P = 0.004$), HLA-mismatch levels were no longer significantly associated with graft loss (Figure 6a and b). Figure 6e and f displays the adjusted Kaplan-Meier survival curves for 10-year DCGS according to each level of match and mismatch from this combined model.

To test the colinearity in our multivariable Cox model, we calculated variance inflation factor. Overall variance inflation factor was 2.223 indicating lack of significant multicollinearity in the model. To further clarify whether matching has a greater effect on graft survival than mismatching does, as proof of principle, we analyzed the subgroup with minimum concordance (Table 1) and compared DCGS between the 0-mismatch, 3-to-5 match cohort with the 0-mismatch, 6-match cohort. Here, we clearly observed an adjusted hazard ratio of 1.18 (95% confidence interval: 1.01–1.45, $P = 0.03$, $n = 8460$) for graft loss at 10 years between 0-mismatch–3- to 5-match transplants compared with 0-mismatch–6-match showing the benefit of 6-match over 0-mismatch (Supplementary Figure S2A). Conversely, we examined the 0-match cohort who could have 3-, 4-, 5- or 6-mismatches in UNOS. Here, the presence of differing mismatch levels within the same match level had no impact on DCGS (adjusted hazard ratio: 0.985; 95% confidence interval: 0.907–1.071; $P = 0.72$) (Supplementary Figure S2B). Together, these data suggest that when adjusted for within the same model exploring the important outcome of graft survival, HLA-matching levels retained an independent association with DCGS, while HLA mismatching lost any independent effects. We also assessed model fitness and strength comparing multivariate models for DCGS as outcome, with either HLA-match or HLA-mismatch using Akaike information criterion (AIC) and Bayesian information criterion (BIC).[16,17] Here we identified lower AIC and BIC (ΔAIC and ΔBIC >10) values for the HLA-match-containing model compared with HLA-mismatch-containing model, showing better model fitness with HLA matching as covariate (Supplementary Table S4).

To explore whether linkage to non-A/B/DR HLA loci plays a role in the apparent preeminent effect of matching over mismatching on graft outcomes, we tabulated matching and mismatching at HLA-DQ loci in the discordant

Table 1 | Matrix distribution according to match and mismatch levels

	Match (%)						
	0	1	2	3	4	5	6
Mismatch							
0	—	—	—	**423 (0.4)**	**1677 (1.7)**	**3533 (3.7)**	**2827 (2.9)**
1	—	—	49 (0.1)	347 (0.4)	1034 (1.1)	1048 (1.1)	—
2	—	59 (0.1)	419 (0.4)	1999 (2.1)	2509 (2.6)	—	—
3	59 (0.1)	689 (0.7)	4496 (4.7)	8279 (8.6)	—	—	—
4	598 (0.6)	6549 (6.8)	17,029 (17.7)	—	—	—	—
5	**4522 (4.7)**	**23,437 (24.4)**	—	—	—	—	—
6	14,654 (15.2)	—	—	—	—	—	—

Bold values represent strata of minimum concordance between human leukocyte antigen match and mismatch levels.

0-mismatch group. HLA-DQ loci reporting in UNOS is independent of A, B, and DR genotypes and are not utilized currently in routine HLA-matching considerations for organ allocation. We excluded all transplants where donors or recipients or both had no HLA-DQ genotyping reported at either loci within the 0-mismatch group (included $n = 3007$). Here, we observed that 0-mismatch, 3-to-5 match transplants had significantly lower proportions of 2-antigen matches at DQ-loci when compared with the 0-mismatch–6-match group, suggesting linkage to HLA-DQ loci (47.0% vs. 57.8%; chi-square $P < 0.0001$) (Supplemental Figure S3).

Sensitivity analyses. The association of HLA-match levels on graft loss and the absence of significant association with HLA-mismatch levels in the combined model were consistent in the cohort after 2003 (Figure 6e and f). Resampling analysis using bootstrapping technique with 10,000 iterations with resubstitution revealed similar results. In univariate analysis, both DGF and 1yr-AR had significant association with DCGS (Table 1). To further examine the mediators of the relationship between HLA matching and graft survival, we examined graft loss in allograft recipients with and without reported DGF or 1yr-AR. In the group without DGF, yet again, we identified that HLA-match levels were significantly associated with 10-year DCGS, while HLA mismatching did not have significant effects ($P = 0.02$ vs. 0.40 respectively) (Supplementary Table S5). Similar independent association of HLA matching and not HLA mismatching on 10-year DCGS was observed in the cohort without 1yr-AR (Supplementary Table S6). These data suggest that while decreased HLA matching influences graft survival through increased risk of DGF or clinical 1yr-AR or both, HLA-match levels also impact DCGS without DGF or 1yr-AR, and these early events may not be the exclusive mediators of this effect.

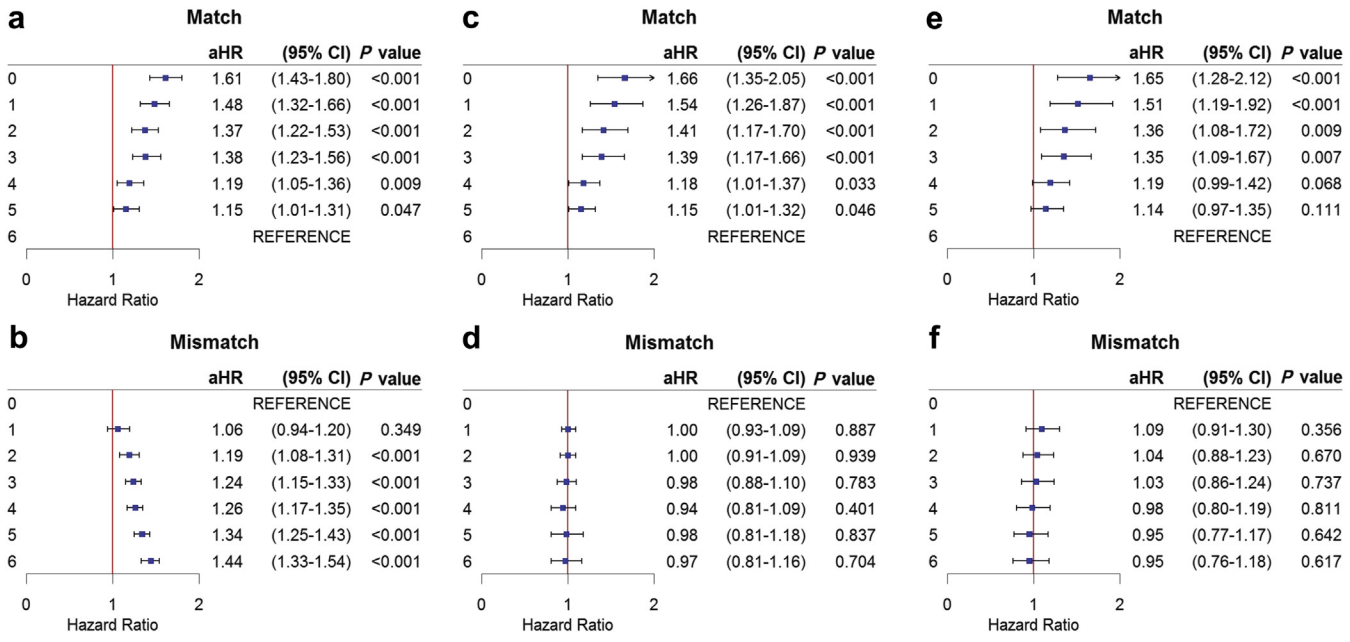


Figure 3 | Human leukocyte antigen (HLA)-matching preferentially impacts delayed graft function when considered in combined models. Figure shows forest plots with hazard ratios evaluating the relationships of delayed graft function with (**a**) HLA-match levels and (**b**) HLA-mismatch levels, in adjusted individual models, and (**c**) HLA-match levels and (**d**) HLA-mismatch levels in adjusted combined models for the whole cohort, as well as the (**e**) 2003 to (**f**) 2012 cohorts. Bullets = mean; whiskers = SD. aHR, adjusted hazard ratio; CI, confidence interval.

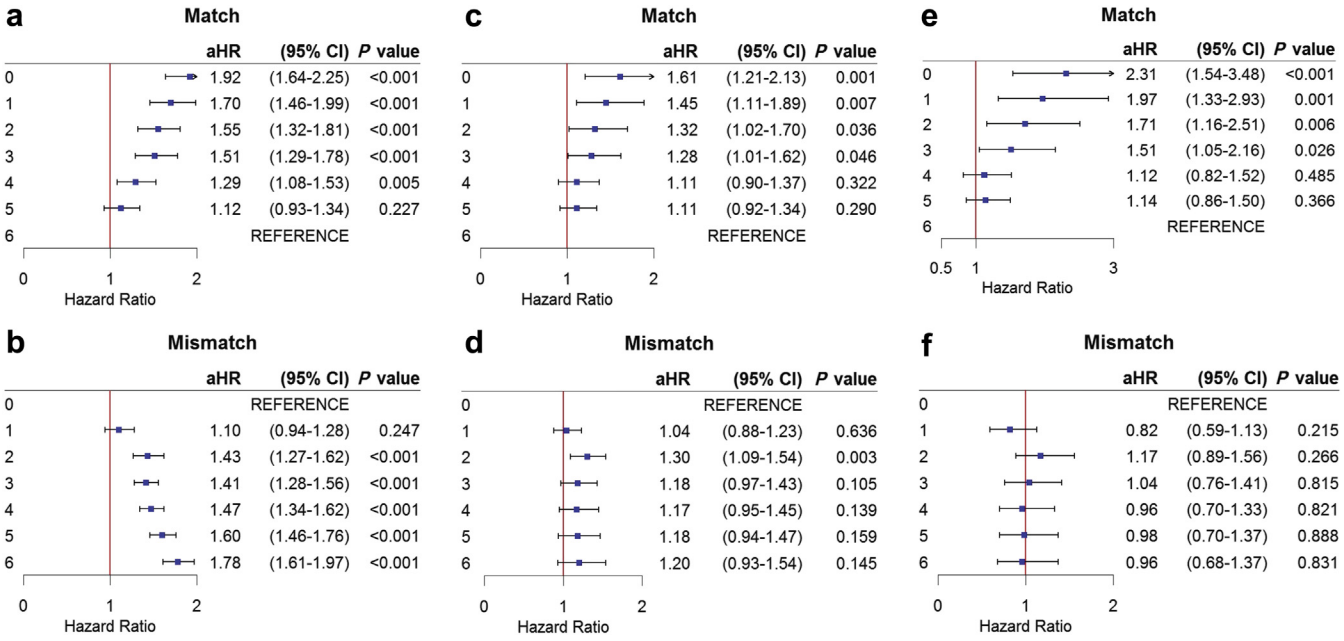


Figure 4| Human leukocyte antigen (HLA)–matching preferentially impacts clinical acute rejection within 1 year posttransplantation when considered in combined models. Figure shows forest plots with hazard ratios evaluating the relationships of clinical acute rejection within 1 year posttransplantation with (**a**) HLA-match levels and (**b**) HLA-mismatch levels, in adjusted individual models, and (**c**) HLA-match levels and (**d**) HLA-mismatch levels in adjusted combined models for the whole cohort, as well as the (**e**) 2003 to (**f**) 2012 cohort. Bullets = mean; whiskers = SD. aHR, adjusted hazard ratio; CI, confidence interval.

DISCUSSION

Here we studied the UNOS/OPTN database of deceased-donor kidney transplants since 1995 to examine the independent association between increased HLA matching and reducing HLA mismatching, on graft outcomes. We reasoned that living related transplants would include identical twin transplants and have greater chances of haplotype identity based on related transplants. Thus, to avoid any confounding that could result from these effects, we excluded all living donor kidney transplants from our analysis. We found that when added to the same adjusted models, HLA matching nearly completely accounted for any effect that has previously been attributed to arise from HLA mismatching.[2–4] This effect was consistent while examining each of the outcomes of 10-year DCGS, 1yr-AR, and DGF. This effect remained apparent in sensitivity and resampling analyses. Our findings from this exploratory analysis suggest that future studies of renal allografts should assess the utility of both HLA matching and HLA mismatching independently as covariates for allograft outcomes, and that HLA matching may be more significantly associated with outcomes.

Numerous analyses from different kidney transplant registries affirm that each level of HLA mismatching continues to have an additive effect on immune events and graft survival.[6,18–20] Notably, we utilized kidney donor risk index as a covariate in our outcome analyses while most prior registry analyses that have reported an effect of HLA mismatching on graft outcomes have not included this variable. In the UNOS database, an allelic mismatch is only reported when the *donor* has an HLA allele at a particular A, B, or DR locus that the *recipient* does not have, and not vice versa. Furthermore, when a genotype is reported as a "blank" for either donor or recipient, this may either be allelic duplication at that locus or less commonly represent an allelic variant that is untypable by the existing polymerase chain reaction primer or probes or both. The reporting of a match on the other hand requires the presence of the same allelic variant at a given locus in both donor and recipient. Newer data suggest that all HLA-mismatches of similar count or at the same locus or both are not equal and that what determines immunogenicity may be final differences in major histocompatibility class II peptides at specific epitopes conferred by different DNA sequences.[21,22] Non-A/B/DR HLA-antigens and non-HLA genetic loci have also been shown to have bearing on allograft outcomes.[11,12,23–25] For instance, our data suggest that A-, B-, and DR-matching may better represent matching at HLA-DQ loci than the extent of mismatches do. HLA-DQ matching has been shown to impact allograft outcomes in several publications.[26,27] Elegant data have also recently demonstrated that non-HLA dependent, innate immune-cell-mediated non-self-recognition and subsequent immune responses have significant impact on graft outcomes.[10,28,29] These immunogenic responses likely involve donor-recipient differences at other genetic loci, not those mapped by HLA-mismatch levels alone. Due to these reasons, we hypothesized that HLA matching would better take into account both genetic linkage to non-A/B/DR loci and could partly reduce class II epitope mismatching. These results from our exploratory analysis in this large dataset substantiate our hypothesis and offer directions for further mechanistic research.

Table 2 | Univariate analysis of covariates with significant effects on 10-years DCGS included in the final model

Variables	n (%)[a]	HR	95% CI	*P* value
Categorical variables				
AA race	42,994 (29)	1.99	1.94–2.04	<0.001
Female sex	58,474 (39.2)	1.003	0.99–1.01	0.626
DM	25,546 (22.7)	1.113	1.06–1.16	<0.001
H/o Dialysis	131,200 (88.6)	1.726	1.64–1.81	<0.001
Transplant date				
1995–1999	34,485 (23.1)	1.402	1.31–1.5	<0.001
2000–2004	39,358 (26.4)	1.26	1.18–1.35	<0.001
2005–2009	49,128 (32.9)	1.156	1.08–1.24	<0.001
2010–2012	26,284 (17.6)	—	—	—
BMI				
<18	4216 (2.9)	—	—	—
18–24.9	53,776 (36.7)	0.965	0.89–1.04	0.370
25–29.9	48,488 (33.1)	0.986	0.911–1.07	0.722
30–34.9	27,398 (18.7)	1.143	1.05–1.24	0.001
≥35	12,534 (8.6)	1.355	1.24–1.48	<0.001
1-year AR	19,606 (18.7)	2.346	2.27–2.42	<0.001
DGF	32,708 (21.9)	2.264	2.2–2.33	<0.001
Thymoglobulin induction	95,465 (63.9)	0.926	0.9–0.95	<0.001
Maint.[b]				
CNI no steroid	20,130 (13.5)	0.497	0.47–0.52	<0.001
CNI with steroid	111,600 (74.8)	0.584	0.56–0.6	<0.001
Numerical variables				
Age	149,255	0.994	0.993–0.995	<0.001
KDRI	145,970	1.916	1.87–1.96	<0.001
CIT	136,853	1.011	1.01–1.012	<0.001
PRA	135,790	1.002	1.001–1.003	<0.001
BMI	146,412	1.007	1.006–1.008	<0.001

AA, African American; AR, acute rejection; BMI, body mass index; CI, confidence interval; CIT, cold ischemia time; CNI, calcineurin inhibitors; DCGS, death-censored graft survival; DGF, delayed graft function; DM, diabetes mellitus; H/o, history of; HR, hazard ratio; KDRI, kidney donor risk index; Maint., maintenance therapy at the time of discharge; PRA, panel reactive antibodies.
[a]Excluding missing data.
[b]Reference group is no CNI therapy.

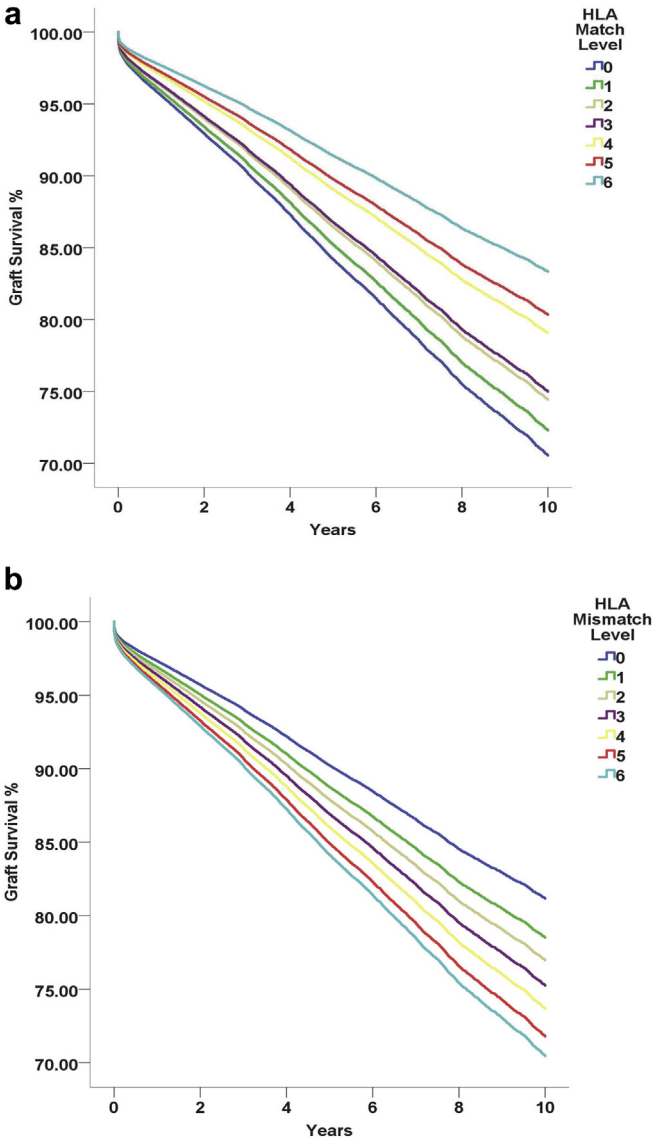


Figure 5 | Impact of human leukocyte antigen (HLA) matching and HLA mismatching on 10 year death-censored graft survival. Lines represent adjusted Kaplan-Meier survival curves for 10-year death-censored graft survival according to (**a**) HLA-match levels and (**b**) HLA-mismatch levels in adjusted individual models.

We observed that the 6-matched transplants represented only one-third of the 0-mismatch group. However, we identified an 18% increased risk for 10-year graft loss in 0-mismatch–3- to 5-match transplants compared to 0-mismatch–6-match transplants showing the benefit of 6-match over 0-mismatch, contradictory to earlier reports with smaller sample sizes.[30] This effect is not explained by identical twins as we excluded living transplants from this analysis and suggests the preeminent effect of match-levels over mismatching. When transplants to African American-recipients were analyzed separately (data not shown), only 194 African Americans had received 6-antigen matched kidneys, representing 0.2% of all and 6.8% of all 6-matched transplants since 1995. This reflects the ethnicity of the deceased donor pool in the United States, because of which any organ-sharing or organ-allocation algorithm in the United States that uses only HLA matching may tend to discriminate against minority ethnicities,[31,32] and changes in organ-sharing algorithms cannot be recommended in the United States based on our preliminary analysis alone. However, first, our findings may have more immediate application in regions with greater similarities between the ethnicities of the donor and recipient pools. Our data should serve as the basis for further studies in the kidney donor risk index/estimated posttransplant survival era, that will test weighted allocation algorithms (for instance, 0-mismatched and 3-, 4-, 5-, 6-antigen matches for different sharing strategies that will incorporate recipient race as a variable) quantifying graft life years gained and wait list times reduced, to obviate the potential for discrimination against minorities with a matching-based strategy in the United States. Additionally, transplant programs preferentially stratify induction therapy or steroid-withdrawal regimens based on 0-mismatch transplants, while studies have reported minimal preferential benefit of such stratification.[33] Other data has suggested significant rates of antibody-mediated and cellular rejection in 0-mismatched deceased-donor transplants, indicating

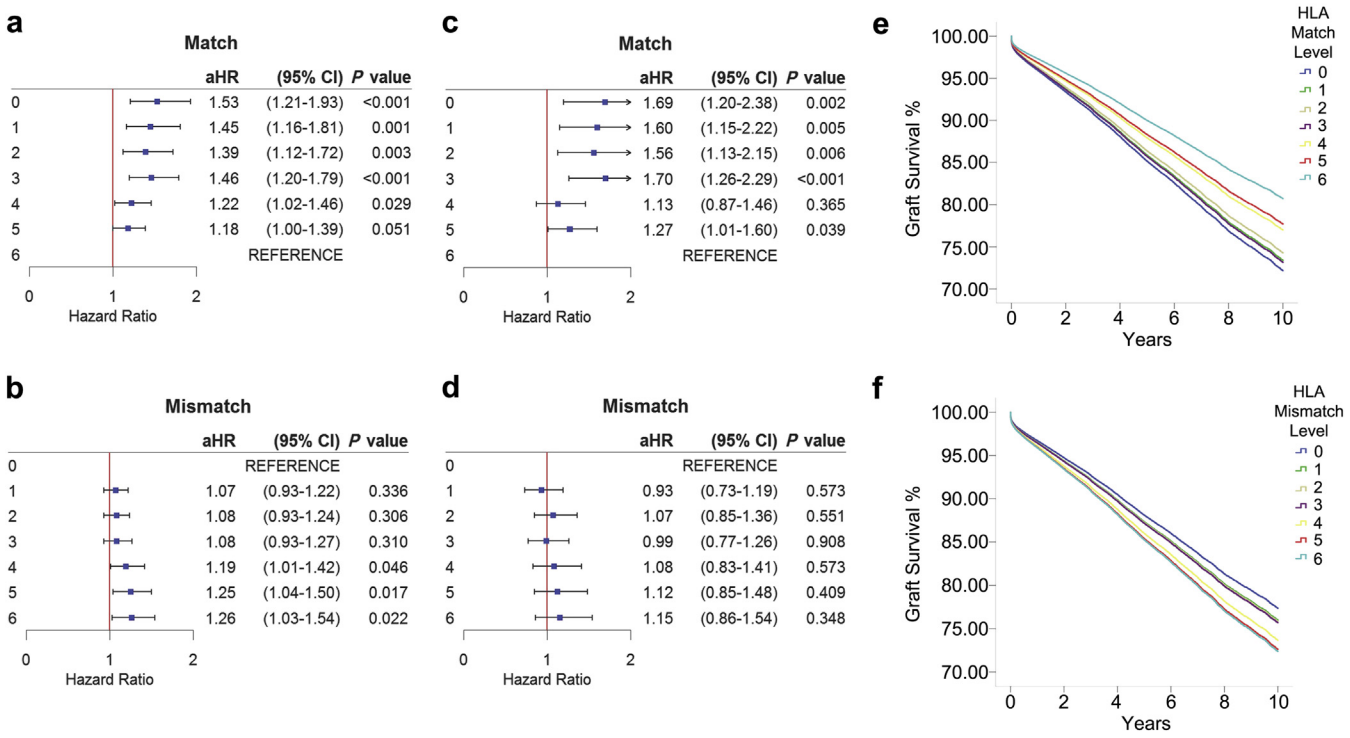


Figure 6 | Human leukocyte antigen (HLA) matching preferentially impacts 10-year death-censored graft survival when considered in combined models. Figure shows forest plots with hazard ratios evaluating the relationships of 10-year death-censored graft survival with (**a**) HLA-match levels and (**b**) HLA-mismatch levels, in adjusted individual models for the whole cohort, and (**c**) HLA-match levels and (**d**) HLA-mismatch levels in adjusted combined models for the 2003 to 2012 cohort. Lines represent adjusted Kaplan-Meier survival curves for 10-year death-censored graft survival for the entire United Network of Organ Sharing cohort according to (**e**) HLA-match levels and (**f**) HLA-mismatch levels in adjusted combined models. Bullets = mean; whiskers = SD. aHR, adjusted hazard ratio; CI, confidence interval.

persisting alloimmunity.[34] Stratifying therapy based on 6-matching could also be a potential application of our findings to be tested prospectively.

Our findings in this retrospective data analysis have limitations. First, this analysis is based on matching-mismatching reports in UNOS data that are derived from 2-digit HLA reports.[9] The incremental discovery of polymorphic A, B, DR loci, as well as the progressive switch from serotyping to genotyping methods across the United States have influenced the labeling of matches and mismatches in UNOS data over the years in our study. Different molecular HLA-typing techniques were used,[35] and serially updated primers or probes or both by vendors may have resulted in heterogeneous reporting of "blanks" at each loci in different laboratories over the years and in turn match and mismatch data. Although we performed stratified analyses for the subgroup after 2003, and adjusted in our multivariable models for year of transplantation, we cannot fully account for any confounding resulting from these technical or nomenclature changes in our analyses. We cannot account for linkage disequilibrium among HLA alleles in this analysis, and some higher matched (3 or more) groups may still represent haplotype similarity, in spite of excluding living related transplants. Some such as A1-B8-DR17 and A2-B44-DR4, which are common in Caucasian ethnicity, could have confounded the analysis.[9] However, we did adjust for donor and recipient races in our analysis to minimize this effect. Subclinical rejection and later AR (i.e., after 1 year) may also contribute to adverse allograft outcome from lower levels of HLA matching, which are not available in UNOS registry; though, we also identified the predominant effect of HLA-matching on DCGS in patients without 1yr-AR. Finally, though the preeminent effect on early and late-allograft outcomes is from HLA matching, rather than mismatching, the absolute effect sizes are small.

In summary, we identified in our exploratory UNOS/OPTN registry analysis of deceased donor transplants that the effect of HLA matching on allograft outcomes nearly completely accounts for effects attributed to HLA mismatching. HLA matching associates with both early events such as DGF and clinical 1yr-AR, as well as 10-year graft survival in models completely adjusted for known predictors including kidney donor risk index. These associations with HLA matching may arise from better predictions of potential linkage to non-HLA genetic loci and possibly better prediction of HLA-epitope matching, and they need further mechanistic and clinical studies to identify potential non-HLA loci of consequence to graft survival. Further analyses using UNOS should preferentially use HLA matching over HLA mismatching as a covariate in analyzing predictive factors for allograft outcomes to improve accuracy. Our current analyses and data from future studies may set the basis to replace HLA mismatching with HLA matching for prioritizing organ allocation.

MATERIALS AND METHODS

Data source

This study used data from the UNOS/OPTN registry. OPTN submits data on all donors, waitlisted candidates, and transplant recipients in the United States. The Health Resources and Services Administration of the US Department of Health and Human Services provide the oversight to the activities of UNOS/OPTN.

Outcome ascertainment

The primary outcome measures were DCGS, 1yr-AR, and (DGF. DGCS was defined as the time from transplantation to graft loss, return to dialysis, or last follow-up with a functioning graft, censored for death or 10 years post–renal transplantation. Transplants where recipients died at any time after graft loss were included as events for the outcome of DCGS at the time of graft loss. DGF was defined as the necessity for dialysis in the first week after surgery, and 1yr-AR was defined as clinical acute rejection within the first year posttransplantation as reported to UNOS.

Study population

All adult, first-time transplant recipients of deceased donor kidney alone (between January 1, 1995, and July 31, 2012), without a history of any organ transplantation, were identified ($n = 151,684$). To avoid confounding from living related transplants, which would show haplotype similarity (3 or more HLA matches) and associated linkage, we excluded all living-donor transplants. Based on *a priori* defined variables, and univariate analysis of factors significantly associated with graft survival, we restricted our outcomes analysis to patients with complete data for both *a priori* identified variables and those associated with DCGS in the UNOS cohort (Figure 2). We did not perform any missing data imputation.

Statistical analysis

Exploratory data analysis. Based on a thorough literature review, we identified factors that have had significant associations with DCGS in prior publications.[36] From the entire dataset of 151,684 patients we identified donor and recipient clinical/demographic variables that were significantly associated with DCGS in univariable analysis (Table 1). Continuous variables were analyzed using analysis of variance, and categorical variables were examined using chi-square. Factors included in the final adjustment for complete data were transplant date; kidney donor risk index; cold ischemia time; recipient's age, sex, race, body mass index, panel reactive antibody, diabetes mellitus, dialysis, HLA matching, HLA mismatching; induction therapy; maintenance therapy at the time of discharge; DGF; and 1yr-AR. Because we used cold ischemia time, panel reactive antibody, induction therapy, and maintenance therapy for adjustment, we adjusted our analysis for the first 14 factors and evaluated the interactions between both of DGF and 1yr-AR with HLA matching and HLA mismatching by stratification. We excluded all transplants where complete data for the 14 covariates included in adjusted analyses was missing, with a final sample size of 96,236 transplants.

We evaluated 1yr-AR in those who were followed for $\geq$1 year posttransplantation. Recipients followed $<$1 year with no documented 1yr-AR, and those who lost their graft in $<$1 year not due to AR were excluded in the analysis, while all those with reported 1yr-AR were included regardless of the follow-up duration. Data was adjusted for the 14 covariates mentioned (excluding DGF and 1yr-AR). Variance inflation factor was calculated to evaluate collinearity.

Model fitness. We utilized AIC and BIC in model selection and the appraisal of model fitness. We first calculated both AIC and BIC in models that include only HLA-match or HLA-mismatch. Then we calculated AIC and BIC for a model that includes all covariates except HLAs, followed by adding HLA-match, or HLA-mismatch, or both. To ascertain model fitness, we computed ΔAIC and ΔBIC and considered the model to better fit if ΔAIC or ΔBIC > 10.[16,17]

Outcomes analysis. We examined the outcomes of DCGS, DGF, and 1yr-AR. DCGS was analyzed using Kaplan-Meier methods, log-rank tests, and Cox proportional hazards models. Logistic regression analysis was used to evaluate DGF and 1yr-AR, treating these as binary outcomes. We separately examined the multivariable adjusted association of HLA-matching and HLA-mismatching levels on 10-year DCGS. Then to determine whether HLA mismatching or HLA matching had a more significant association with our 3 outcomes of interest, we generated a multivariate model including both HLA matching and HLA mismatching as well as the other previously included clinical covariates.

Sensitivity and resampling analysis. Since 2003 when UNOS initiated mandated uniform reporting standards for HLA genotyping laboratories to incorporate progressive use of genotyping in place of serotyping as well as the incremental discovery of new HLA-allelic loci, we analyzed this cohort within the UNOS dataset separately to ensure homogeneity of HLA nomenclature in this cohort. As sensitivity analyses, we examined patients without DGF (outcome 1yr-AR) and those without 1yr-AR (outcome DCGS) in the entire UNOS cohort. We also performed a resampling analysis using bootstrapping technique with a number of samples of 10,000, iterations of 10,000 with resubstitution, and a confidence interval of 95%.

DISCLOSURE

All the authors declared no competing interests.

ACKNOWLEDGMENTS

MCM wishes to acknowledge funding from American Heart Association (AHA 15SDG25870018), Translational Collaborative Research Initiative Grant, Department of Medicine, Icahn School of Medicine at Mount Sinai and philanthropy from Nina and Homer Eaton. RY acknowledges support from the University at Buffalo Innovative Micro-Programs Accelerating Collaboration in Themes (IMPACT), and the National Center for Advancing Translational Sciences of the National Institutes of Health (UL1TR001412 to the University at Buffalo). The authors also wish to acknowledge Arvind Menon, MS, and Cindy Park, CHS, at the Rogosin Immunogenetics Transplantation Center, New York, for technical help.

SUPPLEMENTARY MATERIAL

Figure S1. Distribution of deceased-donor kidney transplants according to human leukocyte antigen (HLA)-matching and HLA-mismatching. The bar graphs show distribution of total deceased donor kidney transplants (1995–2012) by HLA-mismatch levels (white bars) and, HLA-match levels (black bars), each expressed as percentage of total.

Figure S2. Survival curves for 10-year death-censored graft survival (DCGS) of 0-mismatch and 0-match cohorts. Kaplan-Meier survival curves (**A**) comparing 10-year DCGS of 3-, 4-, and 5-match groups ($n = 5633$) with 6-match group ($n = 2827$) within the 0-mismatch cohort (adjusted hazard ratio [aHR]: 1.18; 95% confidence interval [CI]: 1.01–1.45; $P = 0.03$), while (**B**) compares the 10-year DCGS of 3-, 4-, and 5-mismatch groups ($n = 5179$) with 6-mismatch group ($n = 14,654$), within the 0-match cohort (aHR: 0.985; 95% CI: 0.907–1.071; $P = 0.72$).

Figure S3. Number of 0-mismatched transplants with 2-loci-human leukocyte antigen (HLA)-DQ matches. Bar graph compares distribution of the percentage of 0, 1, and 2 loci matches at the HLA-DQ locus between 6-match group ($n = 1352$) and 3-, 4-, 5-match groups ($n = 1655$). Zero-mismatched transplants where either donor or recipient had no HLA-DQ loci genotyping results reported at any loci were excluded from analysis. ***Chi-square $P < 0.0001$.
Table S1A. Clinicodemographic characteristics and match levels.
Table S1B. Clinicodemographic characteristics and mismatch levels.
Table S2. Combined human leukocyte-matching (HLA-M) and human leukocyte-mismatching (HLA-MM) model and risk of clinical acute rejection within 1 year posttransplantation (1yr-AR) in recipients after 2003 without delayed graft function (DGF).
Table S3. Individual human leukocyte-matching (HLA-M) and human leukocyte-mismatching (HLA-MM) and graft loss risk in 2003 to 2012 cohort.
Table S4. Akaike information criterion **(**AIC) and Bayesian information criterion (BIC) assess model fit with HLA matching versus HLA mismatching for the outcome of 10-year death-censored graft survival (DCGS).
Table S5. Combined human leukocyte-matching (HLA-M) and human leukocyte-mismatching (HLA-MM) model and 10 years' graft loss in recipients with and without delayed graft function (DGF).
Table S6. Combined human leukocyte-matching (HLA-M) and human leukocyte-mismatching (HLA-MM) model and 10 years graft loss in recipients with and without clinical acute rejection within 1 year post-transplantation (1ryr-AR).
Supplementary material is linked to the online version of the paper at www.kidney-international.org.

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